

ADVANCED QUANTUM MECHANICS

Fundamentals of quantum mechanics



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LECTURE 3: The hydrogen atom - review of main aspects

Let us focus on the simplest problem that allows clear and instructive analytical solutions, the problem of an electron bound to a proton via Coulomb attraction. The Hydrogen atom is a fundamental problem as it allows to understand spectroscopic quantities and the nature of orbitals. The full understanding of this system will ease the understanding of more complex problems involving many atoms, many electrons or the structure of nucleus, later on.

We will proceed as follows:

- Introduce the single electron problem as a Schrödinger equation for the electron alone (fixing the nucleus in the origin)
- Discuss the conservation of angular momentum and its consequences
- Introduce the angular momentum algebra (spherical harmonics)
- Present the procedure for solving the radial versus angular parts in the position representation
- Present and discuss the wavefunction solutions of the hydrogen problem
- Discuss degeneracies and visualize orbitals

Note that starting with this lecture, I will add a set of online resources that you may find useful for visualizing and exploring the discussed concepts [1, 2, 3, 4, 5, 6].

Stating the problem: an electron and a proton

Let us write the total Hamiltonian for an electron bound to a proton via Coulomb attraction:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + \frac{\hat{\mathbf{P}}^2}{2M} - \frac{e^2}{4\pi\epsilon_0 \|\hat{\mathbf{r}} - \hat{\mathbf{R}}\|}.$$

This is already a two-body problem described in the position representation by a two-particle wavefunction $\psi(\mathbf{R}, \mathbf{r})$. To deduce this, it would require us to solve a stationary Schrödinger equation

$$H \psi(\mathbf{R}, \mathbf{r}) = E \psi(\mathbf{R}, \mathbf{r}),$$

where H is the position representation form of $\hat{H}$.

However, a direct simplification is possible, based on the fact that the mass of the proton is much larger than that of the electron. In atomic mass units $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$, (which is defined as $1/12$ of the mass of a neutral ^{12}C atom in its ground state), we have for the proton mass $M = 1.007276 \text{ u}$ and for the electron mass $m = 0.00054858 \text{ u}$, leading to the ratio

$$\frac{M}{m} \approx 1836.15.$$

The "heavy nucleus" observation suggests fixing the proton at the origin and reducing our treatment to a single electron

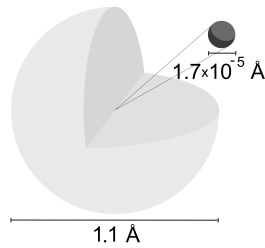


Figure 1: Comparison of electronic orbital size versus size of nucleus.

described by the operators $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$, with Hamiltonian

$$\hat{H} = \hat{H}_0 + V(\hat{r}) = \frac{\hat{\mathbf{p}}^2}{2m} - \frac{e^2}{4\pi\epsilon_0 \hat{r}}.$$

We keep m for the mass of the electron; this actually is the reduced mass

$$\frac{mM}{m+M},$$

but very close to the electron mass. Here the radial position operator is defined as

$$\hat{r} \equiv \|\hat{\mathbf{r}}\| = \sqrt{\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}}.$$

For the curious reader, go ahead and figure out what it means to define the square root of an operator :).

Single particle versus many-body quantum physics

Distinguishable particle - In later lectures, we will come to more complex many-body problems. For example, we will consider a very simple model for a molecule: the ionized hydrogen molecule H_2^+ with two protons with coordinates $\mathbf{R}_1$ and $\mathbf{R}_2$ and one electron with coordinate $\mathbf{r}$. The wavefunction to be found there is

$$\psi(\mathbf{R}_1, \mathbf{R}_2, \mathbf{r}).$$

This is a many-body system, but first, electrons and protons are distinguishable. In addition, the protons are fixed pretty well in their position, not allowing for an overlap of their position, which makes the problem separable as well. The solution involves first the introduction of the relative motion coordinate $\mathbf{R} = \mathbf{R}_2 - \mathbf{R}_1$ and the simplification of the desired wavefunction to $\psi(\mathbf{R}, \mathbf{r})$. A Born-Oppenheimer approximation is then performed, which will allow us to solve first the electronic problem for fixed nuclear separation and then the nuclear motion on the resulting potential energy surface.

Indistinguishable particles - A further level of complexity appears in the Helium atom, where two electrons orbit a (fixed) nucleus placed at the origin. The wavefunction then depends on two electronic position operators,

$$\psi(\mathbf{r}_1, \mathbf{r}_2),$$

and now the particles are indistinguishable fermions which share the space. The wavefunction must be anti-symmetric under the exchange of $\mathbf{r}_1 \xleftrightarrow{\text{exchange}} \mathbf{r}_2$ introducing quantum statistics as a fundamental constraint.

Consequences of the spherically symmetric potential

We focus on a single degree of freedom, the motion of the electron, and drop the color coding as it is not necessary any longer. Now we make use of the fact that we have a spherically symmetric potential. This suggests the use of the angular

momentum operators

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = \begin{pmatrix} \hat{y}\hat{p}_z - \hat{z}\hat{p}_y \\ \hat{z}\hat{p}_x - \hat{x}\hat{p}_z \\ \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \end{pmatrix},$$

or in simplified notation $\hat{L}_i = \varepsilon_{ijk} \hat{x}_j \hat{p}_k$. The following commutators are satisfied

$$[\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k,$$

where ε_{ijk} is the [Levi-Civita symbol](#). One then defines

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = (\hat{y}\hat{p}_z - \hat{z}\hat{p}_y)^2 + (\hat{z}\hat{p}_x - \hat{x}\hat{p}_z)^2 + (\hat{x}\hat{p}_y - \hat{y}\hat{p}_x)^2.$$

One can check the conservation of angular momentum ([Noether's theorem](#) implies this from the rotational symmetry)

$$[\hat{H}, \hat{L}^2] = 0, \quad [\hat{H}, \hat{L}_z] = 0,$$

meaning that the Hamiltonian and the operators $\hat{L}^2$ and $\hat{L}_z$ admit common eigenstates. This will help us in constructing and understanding electronic orbitals in terms of spherical harmonics. [The proof for these commutations is left for Exercise 4.](#)

Since we are simultaneously diagonalizing three operators, we introduce a quantum number for each of them and denote the eigenstates as $|n\ell m\rangle$ writing the eigenvalue problem as

$$\hat{H} |n, \ell, m\rangle = E_{n, \ell, m} |n, \ell, m\rangle.$$

we will see later what are the possible values of the three numbers and also we will find out that there is a full degeneracy for ℓ and m as the energy will only depend on the value of n .

Eigenbasis of $\hat{L}^2$ and $\hat{L}_z$

We describe in the [Appendix 1](#) a procedure to find the eigenvalues and eigenstates of the angular momentum operator. Here we restrict the presentation only to the important results.

Angular momentum eigenstates

The simultaneous eigenkets of the angular momentum operators are

$$\hat{L}^2 |\ell, m\rangle = \hbar^2 \ell(\ell+1) |\ell, m\rangle, \quad \hat{L}_z |\ell, m\rangle = \hbar m |\ell, m\rangle,$$

with quantum numbers

$$\ell = 0, 1, 2, \dots, \quad m = -\ell, -\ell+1, \dots, \ell.$$

These states form an **orthonormal** and **complete** basis:

$$\langle \ell', m' | \ell, m \rangle = \delta_{\ell\ell'} \delta_{mm'}, \quad \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} |\ell, m\rangle \langle \ell, m| = \mathbb{1}.$$

The ladder operators are defined as

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y,$$

and act within a fixed- ℓ subspace as

$$\hat{L}_{\pm} |\ell, m\rangle = \hbar \sqrt{\ell(\ell+1) - m(m \pm 1)} |\ell, m \pm 1\rangle.$$

They raise or lower the magnetic quantum number m by one unit without changing ℓ .

Spherical harmonics

Let us ask how this operator looks like in the position representation. One can use the procedure detailed in the previous lecture to find the matrix elements of any operator in the position representation. In short

$$\begin{aligned}\langle \mathbf{r} | \hat{\mathbf{L}}^2 | \mathbf{r}' \rangle &= \langle \mathbf{r} | \hat{L}_x^2 | \mathbf{r}' \rangle + \langle \mathbf{r} | \hat{L}_y^2 | \mathbf{r}' \rangle + \langle \mathbf{r} | \hat{L}_z^2 | \mathbf{r}' \rangle = \langle \mathbf{r} | \hat{L}_x \hat{1} \hat{L}_x | \mathbf{r}' \rangle + \langle \mathbf{r} | \hat{L}_y \hat{1} \hat{L}_y | \mathbf{r}' \rangle + \langle \mathbf{r} | \hat{L}_z \hat{1} \hat{L}_z | \mathbf{r}' \rangle \\ &= \int d^3 r_1 \langle \mathbf{r} | \hat{L}_x | \mathbf{r}_1 \rangle \langle \mathbf{r}_1 | \hat{L}_x | \mathbf{r}' \rangle + \int d^3 r_1 \langle \mathbf{r} | \hat{L}_y | \mathbf{r}_1 \rangle \langle \mathbf{r}_1 | \hat{L}_y | \mathbf{r}' \rangle + \int d^3 r_1 \langle \mathbf{r} | \hat{L}_z | \mathbf{r}_1 \rangle \langle \mathbf{r}_1 | \hat{L}_z | \mathbf{r}' \rangle.\end{aligned}$$

From here, one needs to compute the matrix elements of each component. For example

$$\langle \mathbf{r} | \hat{L}_x | \mathbf{r}_1 \rangle = \langle \mathbf{r} | (\hat{y} \hat{p}_z - \hat{z} \hat{p}_y) | \mathbf{r}_1 \rangle = \int d^3 r_2 \left[\langle \mathbf{r} | \hat{y} | \mathbf{r}_2 \rangle \langle \mathbf{r}_2 | \hat{p}_z | \mathbf{r}_1 \rangle - \langle \mathbf{r} | \hat{z} | \mathbf{r}_2 \rangle \langle \mathbf{r}_2 | \hat{p}_y | \mathbf{r}_1 \rangle \right] = -i\hbar(y \partial_z - z \partial_y) \delta(\mathbf{r} - \mathbf{r}_1).$$

Following this kind of reasoning and finally transforming to [spherical coordinates](#), one ends up with

$$\langle \mathbf{r} | \hat{\mathbf{L}}^2 | \mathbf{r}' \rangle = -\hbar^2 \underbrace{\left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right)}_{\text{angular part denoted by } -\Delta_{(\theta, \phi)}} \delta^{(3)}(\mathbf{r} - \mathbf{r}'),$$

where

$$\delta^{(3)}(\mathbf{r} - \mathbf{r}') = \frac{1}{r^2} \delta(r - r') \delta(\cos \theta - \cos \theta') \delta(\phi - \phi').$$

Similarly, one can easily show that

$$\langle \mathbf{r} | \hat{L}_z | \mathbf{r}' \rangle = -i\hbar \frac{\partial}{\partial \phi} \delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

Properties of spherical harmonics

The eigenfunctions in the position representation are the [spherical harmonics](#)

$$Y_{\ell m}(\theta, \phi) = \langle \theta, \phi | \ell, m \rangle,$$

which satisfy

$$\Delta_{(\theta, \phi)} Y_{\ell m}(\theta, \phi) = \ell(\ell + 1) Y_{\ell m}(\theta, \phi), \quad \frac{\partial}{\partial \phi} Y_{\ell m}(\theta, \phi) = -im Y_{\ell m}(\theta, \phi).$$

Generally, they are connected to the [associated Legendre polynomials](#)

$$Y_{\ell m}(\theta, \phi) = (-1)^m \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - m)!}{(\ell + m)!}} P_{\ell}^m(\cos \theta) e^{im\phi}.$$

The functions are orthonormal

$$\int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\phi Y_{\ell m}^*(\theta, \phi) Y_{\ell' m'}(\theta, \phi) = \delta_{\ell \ell'} \delta_{m m'},$$

and complete

$$\sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} Y_{\ell m}^*(\theta, \phi) Y_{\ell m}(\theta', \phi') = \delta(\cos \theta - \cos \theta') \delta(\phi - \phi'),$$

over the range $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$ To get a better grasp, here are some examples

$$Y_{00} = \frac{1}{\sqrt{4\pi}}, \quad Y_{1, \pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}, \quad Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos \theta.$$

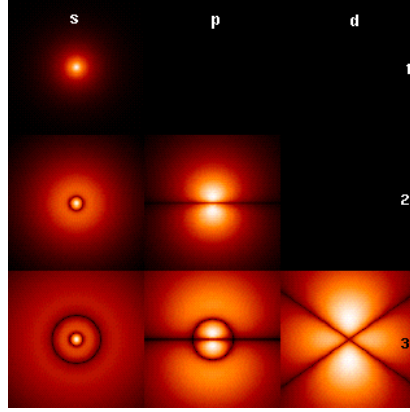


Figure 2: Density plot of the electronic orbitals showcasing the orbital dependence for $\ell = 0(s)$, $\ell = 1(p)$ and $\ell = 2(d)$.

Solving the Schrödinger equation for the Hydrogen atom in the position representation

In the position representation the stationary Schrödinger equation for hydrogen in spherical coordinates is

$$\left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \right] \psi(r, \theta, \phi) = E \psi(r, \theta, \phi).$$

Because the potential depends only on r , we pass to spherical coordinates (r, θ, ϕ) . It is convenient to separate the Laplacian into a radial and an angular part:

$$\nabla^2 = \underbrace{\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right)}_{\text{radial part}} + \frac{1}{r^2} \Delta_{(\theta, \phi)}.$$

We look for separable solutions of the following form

$$\psi(\mathbf{r}) = R(r) f(\theta, \phi).$$

Acting with the Laplacian gives

$$\nabla^2 [R(r) f(\theta, \phi)] = f(\theta, \phi) \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR(r)}{dr} \right) + \frac{R(r)}{r^2} \Delta_{(\theta, \phi)} f(\theta, \phi).$$

Inserting this into the Schrödinger equation, dividing by $R(r) f(\theta, \phi)$ and multiplying through by r^2 we obtain

$$\underbrace{-\frac{\hbar^2}{2m} \frac{1}{R(r)} \frac{d}{dr} \left(r^2 \frac{dR(r)}{dr} \right) - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} - E r^2}_{\text{depends only on } r} = \underbrace{\frac{\hbar^2}{2m} \frac{1}{f(\theta, \phi)} \Delta_{(\theta, \phi)} f(\theta, \phi)}_{\text{depends only on } \theta, \phi}.$$

This cleanly separates the radial and angular dependence, meaning that both must equal the same constant (as the equality holds for any values of the three variables).

This means we have to first solve the eigenvalue problem of $\Delta_{(\theta, \phi)}$, which we already did! We already found the spherical harmonics as solutions $\Delta_{(\theta, \phi)} Y_{\ell, m} = \ell(\ell + 1) Y_{\ell, m}$ so we can identify f and go ahead and solve the radial part

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\hbar^2 \ell(\ell + 1)}{2m r^2} R(r) - \frac{e^2}{4\pi\epsilon_0 r} R(r) = E R(r).$$

This is a second order differential equation with known solutions - the associated Laguerre polynomials. One should go through the steps of solving it at least once in a lifetime. However we assume here that you already did or will go through it using one of the listed textbooks for basic quantum mechanics or the sketch in Appendix 2.

Here we list here the eigenvalues and eigenfunctions. First we introduce the Bohr radius (with $\alpha = 1/137$ denoting the fine structure constant)

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m e^2} = \frac{\hbar}{m c \alpha} \approx 5.29 \times 10^{-11} \text{ m} = 0.529 \text{ \AA}.$$

Energy levels and radial wavefunctions

The energies of the electron in the hydrogen atom are

$$E_n = -\frac{\hbar^2}{2m a_0^2} \frac{1}{n^2}, \quad n = 1, 2, 3, \dots$$

Notice that one connect the ground state to the rest energy of electron via the fine structure constant

$$E_1 = -\frac{1}{2} m c^2 \alpha^2.$$

The corresponding radial wavefunctions are

$$R_{n\ell}(r) = \underbrace{\left(\frac{2}{na_0}\right)^{3/2} \sqrt{\frac{(n-\ell-1)!}{2n(n+\ell)!}}}_{\text{normalization}} \underbrace{\left(\frac{2r}{na_0}\right)^\ell}_{\text{small-}r \text{ behavior}} \underbrace{e^{-\frac{r}{na_0}}}_{\text{large-}r \text{ exponential decay}} \underbrace{L_{n-\ell-1}^{2\ell+1}\left(\frac{2r}{na_0}\right)}_{\text{associated Laguerre polynomial}}.$$

Here are the first two radial wavefunctions

$$R_{10}(r) = 2 a_0^{-3/2} e^{-r/a_0}, \quad R_{20}(r) = \frac{1}{2\sqrt{2} a_0^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-r/(2a_0)}.$$

To understand what the prediction of these analytical solutions are, we refer to Fig. 3, showing energy level sequence, the degeneracy and the standard notation for transitions lines and orbitals.

In the next lecture we will list all the mechanisms that lift this degeneracy and will perform some calculations using what we will introduce as perturbation theory.

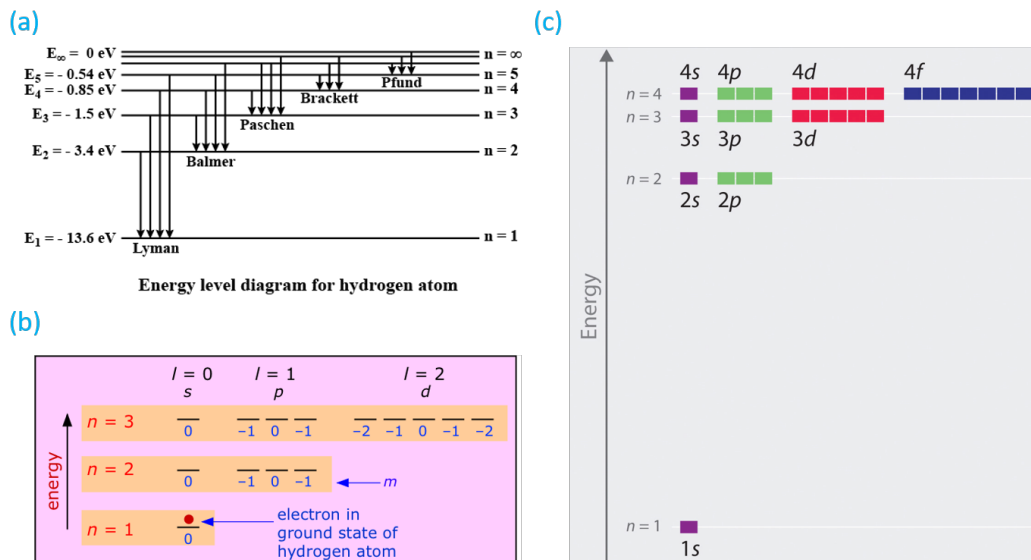


Figure 3: (a) The nomenclature of spectroscopic series. (b) Three energy levels with their degeneracy. (c) Same but including the case $\ell = 3$.

References

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1 Appendix: Constructing the common eigenstates of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$

Start with the angular momentum ladder operators

$$\hat{L}_{\pm} := \hat{L}_x \pm i\hat{L}_y, \quad [\hat{L}_z, \hat{L}_{\pm}] = \pm\hbar\hat{L}_{\pm}, \quad [\hat{L}_+, \hat{L}_-] = 2\hbar\hat{L}_z.$$

Using the definitions of $\hat{L}_{\pm}$ one can check that

$$\hat{\mathbf{L}}^2 = \hat{L}_- \hat{L}_+ + \hat{L}_z^2 + \hbar\hat{L}_z = \hat{L}_+ \hat{L}_- + \hat{L}_z^2 - \hbar\hat{L}_z,$$

and importantly,

$$[\hat{\mathbf{L}}^2, \hat{L}_z] = 0, \quad [\hat{\mathbf{L}}^2, \hat{L}_{\pm}] = 0.$$

Thus, we can choose simultaneous eigenstates of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$:

$$\hat{\mathbf{L}}^2 |\ell, m\rangle = \lambda_{\ell} |\ell, m\rangle, \quad \hat{L}_z |\ell, m\rangle = \lambda_m |\ell, m\rangle.$$

Applying $[\hat{L}_z, \hat{L}_{\pm}]$ to $|\ell, m\rangle$ gives:

$$\hat{L}_z(\hat{L}_{\pm} |\ell, m\rangle) = (\lambda_m \pm \hbar) (\hat{L}_{\pm} |\ell, m\rangle). \quad (1)$$

We define the minimal value where $\hat{L}_- |\ell, m_{\min}\rangle = 0$. Then we apply $\hat{L}_+$ to get Thus $\hat{L}_{\pm} |\ell, m\rangle$ is either zero or an eigenstate of $\hat{L}_z$ with eigenvalue $\lambda_m \pm \hbar$. Therefore,

$$\hat{L}_{\pm} |\ell, m\rangle = C_{\ell, m}^{(\pm)} |\ell, m \pm 1\rangle, \quad (2)$$

showing that the eigenvalues of $\hat{L}_z$ form a uniformly spaced ladder with spacing $\hbar$. We therefore write

$$\lambda_m = \hbar m, \quad (3)$$

where m is an integer.

Next, we use the operator identity

$$\hat{\mathbf{L}}^2 = \hat{L}_- \hat{L}_+ + \hat{L}_z^2 + \hbar\hat{L}_z. \quad (4)$$

Consider a highest weight state $|\ell, m_{\max}\rangle$ defined by

$$\hat{L}_+ |\ell, m_{\max}\rangle = 0. \quad (5)$$

Applying (4) to this state gives:

$$\hat{\mathbf{L}}^2 |\ell, m_{\max}\rangle = \left(\hat{L}_z^2 + \hbar \hat{L}_z \right) |\ell, m_{\max}\rangle = \hbar^2 m_{\max}(m_{\max} + 1) |\ell, m_{\max}\rangle. \quad (6)$$

Thus,

$$\lambda_\ell = \hbar^2 m_{\max}(m_{\max} + 1). \quad (7)$$

We now *define*

$$m_{\max} \equiv \ell, \quad (8)$$

so that

$$\lambda_\ell = \hbar^2 \ell(\ell + 1). \quad (9)$$

Repeating the argument for the lowest weight state gives

$$m_{\min} = -\ell. \quad (10)$$

Since the ladder steps change m by integers, the allowed values of m are

$$m = -\ell, -\ell + 1, \dots, \ell - 1, \ell. \quad (11)$$

2 Appendix: Solving the radial part

Let's list the equation again:

$$-\frac{\hbar^2}{2\mu} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\hbar^2 \ell(\ell + 1)}{2\mu r^2} R(r) - \frac{e^2}{4\pi\epsilon_0 r} R(r) = E R(r).$$

We define

$$u(r) = r R(r) \quad \Rightarrow \quad u(0) = 0,$$

which transforms the radial equation into

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left[\frac{\hbar^2 \ell(\ell + 1)}{2\mu r^2} - \frac{e^2}{4\pi\epsilon_0 r} \right] u(r) = E u(r).$$

Small r behavior. For $r \rightarrow 0$ the $\frac{1}{r^2}$ term dominates:

$$-\frac{\hbar^2}{2\mu} u''(r) + \frac{\hbar^2 \ell(\ell + 1)}{2\mu r^2} u(r) \approx 0.$$

Try $u(r) \sim r^s$:

$$s(s - 1) = \ell(\ell + 1) \quad \Rightarrow \quad s = \ell + 1 \text{ or } s = -\ell.$$

Regularity at $r = 0$ excludes $r^{-\ell}$, hence

$$\boxed{u(r) \sim r^{\ell+1} \quad (r \rightarrow 0).}$$

Large r behavior. For $r \rightarrow \infty$, the $1/r$ and $1/r^2$ terms vanish:

$$-\frac{\hbar^2}{2\mu} u''(r) \approx E u(r), \quad E < 0.$$

Define

$$\kappa = \sqrt{-\frac{2\mu E}{\hbar^2}} > 0,$$

then

$$u''(r) = \kappa^2 u(r) \Rightarrow u(r) = Ae^{\kappa r} + Be^{-\kappa r}.$$

Normalizability requires $A = 0$, so

$$u(r) \sim e^{-\kappa r} \quad (r \rightarrow \infty).$$

Introduce the Bohr radius (with reduced mass μ)

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2},$$

and the dimensionless variable

$$\rho = \frac{2r}{na_0}.$$

Motivated ansatz:

$$u(r) = \rho^{\ell+1} e^{-\rho/2} f(\rho).$$

Substituting into the equation yields the associated Laguerre differential equation:

$$\rho f''(\rho) + (2\ell + 2 - \rho)f'(\rho) + (n - \ell - 1)f(\rho) = 0.$$

Normalizability requires the series to terminate, so $n - \ell - 1$ must be a nonnegative integer. Therefore,

$$f(\rho) = L_{n-\ell-1}^{2\ell+1}(\rho),$$

where L_k^α is an associated Laguerre polynomial. This condition determines the allowed bound-state energies:

$$E_n = -\frac{\hbar^2}{2\mu a_0^2} \frac{1}{n^2} = -\frac{\mu e^4}{2(4\pi\epsilon_0)^2 \hbar^2} \frac{1}{n^2}, \quad n = 1, 2, \dots$$

Thus the normalized radial wavefunctions are

$$R_{n\ell}(r) = \left(\frac{2}{na_0}\right)^{3/2} \sqrt{\frac{(n-\ell-1)!}{2n(n+\ell)!}} e^{-r/(na_0)} \left(\frac{2r}{na_0}\right)^\ell L_{n-\ell-1}^{2\ell+1}\left(\frac{2r}{na_0}\right).$$